

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	30 September 2026
Team ID	SWTID-2026-8273
Project Name	Implement Client Script & UI Policy (Incident)
Maximum Marks	4 Marks

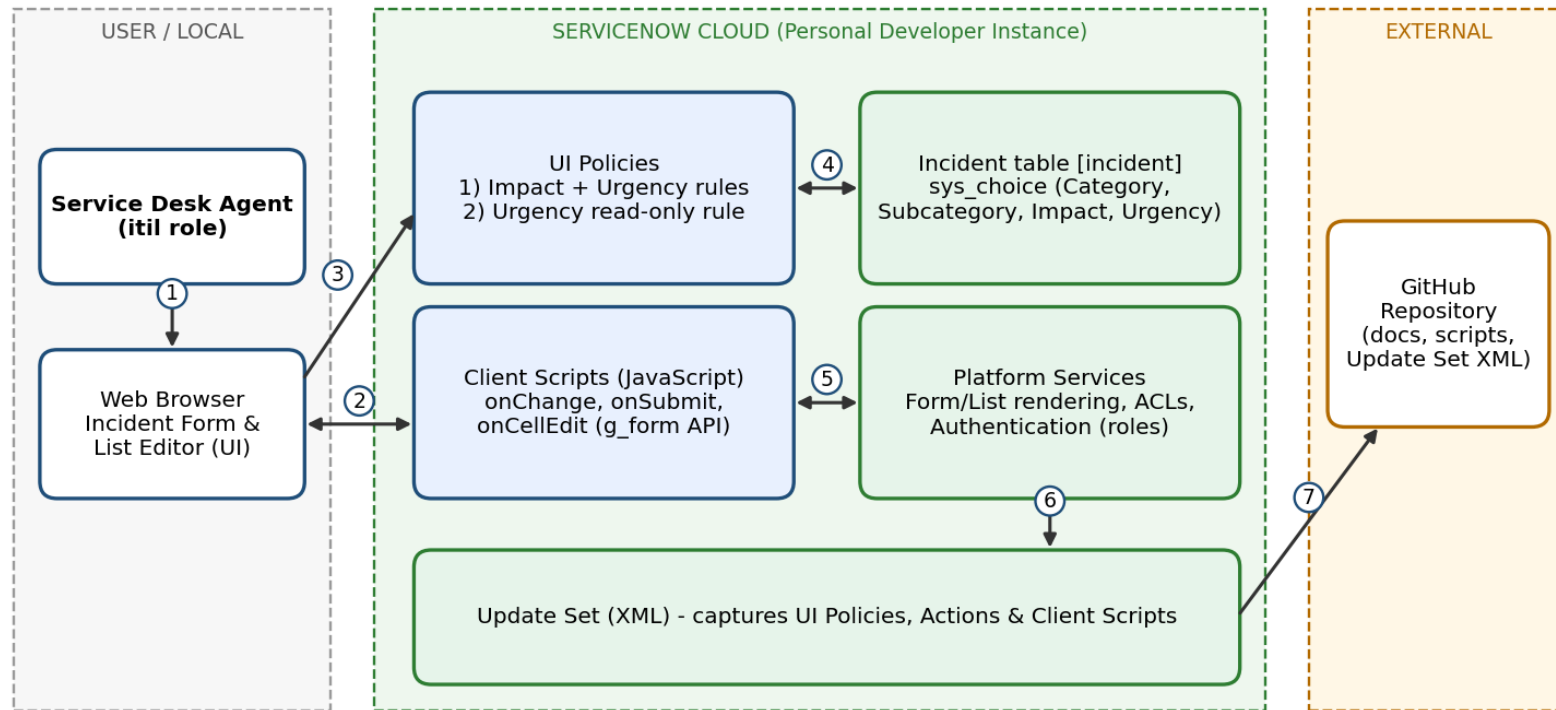
Technical Architecture:

The deliverable includes the architectural diagram below and the information as per Table-1 and Table-2.

Example: Dynamic field control and validation on the ServiceNow Incident form

Reference: <https://developer.servicenow.com>

Technical Architecture: Client Script & UI Policy (Incident) on ServiceNow



1 User works in browser 2 Form/list events trigger client scripts 3 UI Policy evaluated on load/change 4 Policies read/write field state 5 Scripts run against form/list 6 Changes captured 7 Commit to GitHub

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	How the agent interacts with the application: Incident form and Incident list (list editor) in the browser	ServiceNow Platform UI (Next Experience / UI16 forms and lists), HTML / CSS rendered by the platform
2.	Application Logic-1	Declarative field control: mandatory / visible based on Impact and Urgency; Urgency read-only on Resolved / Closed	ServiceNow UI Policies and UI Policy Actions
3.	Application Logic-2	Dynamic behavior: onChange (Category / Subcategory), onSubmit (validation), onCellEdit (list-editor validation)	Client Scripts in JavaScript using the GlideForm API (g_form)

S.No	Component	Description	Technology
4.	Application Logic-3	Packaging and promotion of all configuration between instances	ServiceNow Update Sets (XML)
5.	Database	Incident records, choice lists (Category, Subcategory, Impact, Urgency) and configuration records (sys_ui_policy, sys_script_client)	ServiceNow relational database (Incident [incident], sys_choice tables)
6.	Cloud Database	Managed database service hosting the instance data	ServiceNow Now Platform instance database (Personal Developer Instance)
7.	File Storage	Attachments on incidents; documentation and Update Set XML stored in version control	ServiceNow attachments (sys_attachment); GitHub repository
8.	External API-1	Not required; the solution runs entirely on the platform	N/A
9.	External API-2	Source control for documentation, scripts and Update Set XML	GitHub
10.	Machine Learning Model	Not applicable to this project	N/A
11.	Infrastructure (Server / Cloud)	Application deployment on cloud	Local Server Configuration: none (browser-based access only) Cloud Server Configuration: ServiceNow Personal Developer Instance (PDI) hosted on ServiceNow cloud

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	No open-source framework is required; logic uses the platform's client APIs and standard JavaScript. Git is used for version control.	GlideForm API (g_form), JavaScript, Git / GitHub
2.	Security Implementations	Role-based access control and authenticated sessions; HTTPS/TLS; client-side rules treated as usability controls, with Business Rules / ACLs recommended for server-side enforcement	ServiceNow roles and ACLs, HTTPS/TLS, Business Rules (recommended)
3.	Scalable Architecture	Multi-tier platform (browser, application nodes, database); rules are declarative, condition-driven and reusable across incidents	ServiceNow Now Platform multi-tier architecture

S.No	Characteristics	Description	Technology
4.	Availability	Client logic runs in the browser and follows the instance's availability. Production ServiceNow instances run on redundant cloud infrastructure; the PDI is for development and testing only	ServiceNow cloud hosting
5.	Performance	Logic executes in the browser with no server round trips; UI Policies preferred over scripts; no synchronous GlideRecord calls; isLoading guards prevent needless processing	UI Policies, asynchronous client patterns

References:

<https://developer.servicenow.com>

<https://docs.servicenow.com>

<https://c4model.com/>